

HAMRA, Sweden

WMO: 023290

Lat: 61.6599N

Lon: 14.9918E

Elev: 1493

StdP: 13.92

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	2.5	-6.6	4.1	-0.2	-2.5	5.1	4.3	17.4	32.3	15.6	31.6	4.9	350	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	76.1	60.6	72.7	59.4	69.3	57.9	63.3	71.2	61.5	68.9	59.7	66.5	5.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.4	82.9	65.3	58.5	77.3	64.1	56.7	72.4	62.5	29.4	71.3	28.1	68.7	26.8	66.5	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	12.4	11.2	-6.7	80.4	4.5	2.4	-9.9	82.1	-12.5	83.5	-15.0	84.9	-18.3	86.6		
			-6.7	65.1	4.2	2.0	-9.8	66.6	-12.2	67.8	-14.6	68.9	-17.6	70.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	21.4	22.4	28.5	37.7	46.9	54.1	58.9	56.1	48.4	38.1	29.4	24.2
	DBStd	15.09	8.92	9.43	8.08	6.79	7.37	6.28	5.70	5.50	5.73	6.77	7.55	9.27
	HDD50	4775	885	774	668	374	153	26	2	10	95	371	617	800
	HDD65	9540	1350	1194	1133	820	562	331	207	281	497	833	1067	1265
	CDD50	737	0	0	0	4	56	150	277	199	48	3	0	0
	CDD65	27	0	0	0	0	0	5	17	5	0	0	0	0
	CDH74	197	0	0	0	0	6	44	124	23	0	0	0	0
	CDH80	17	0	0	0	0	0	2	14	1	0	0	0	0
Wind	WSAvg	5.4	5.6	5.7	6.2	5.9	5.7	5.1	4.6	4.5	4.9	5.6	5.7	5.5
Precipitation	PrecAvg	28.2	1.7	1.2	1.3	1.7	2.1	3.2	3.8	3.7	2.7	2.6	2.3	1.8
	PrecMax	42.1	3.9	2.7	2.8	5.9	4.4	6.0	9.5	6.6	5.9	6.8	6.9	3.0
	PrecMin	20.5	0.3	0.2	0.1	0.1	0.4	0.6	0.5	1.8	0.7	0.5	0.7	0.2
	PrecStd	4.7	0.9	0.6	0.7	1.4	1.0	1.2	2.1	1.4	1.4	1.5	1.4	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	43.3	51.2	62.8	74.7	78.8	82.0	76.9	67.5	57.7	48.8	43.7
		MCWB	37.0	36.9	41.0	48.3	56.2	59.1	62.6	63.0	56.4	51.6	46.3	40.5
	2%	DB	37.9	40.1	46.3	58.3	69.6	74.4	77.6	72.6	63.6	53.0	44.8	40.5
		MCWB	35.2	35.6	39.0	45.9	54.0	58.2	62.5	61.3	55.0	48.7	43.0	37.7
	5%	DB	35.7	37.3	42.6	53.5	64.9	70.5	73.7	69.1	60.3	50.2	41.7	38.2
		MCWB	33.7	34.0	36.5	43.2	52.2	56.8	60.5	59.4	53.7	46.5	40.3	35.8
	10%	DB	33.2	34.6	39.6	49.2	60.0	65.8	69.8	65.6	57.2	47.6	39.2	35.8
		MCWB	31.4	31.9	34.7	40.6	49.7	54.5	59.3	57.9	51.9	44.8	38.0	33.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	38.5	42.6	50.7	58.4	63.6	65.6	64.7	58.3	52.9	47.2	41.3
		MCDB	40.0	41.8	49.5	61.1	69.1	72.6	75.9	71.9	63.9	55.7	48.3	42.7
	2%	WB	35.6	36.2	39.5	46.7	55.8	59.6	63.7	62.8	56.2	49.7	43.1	38.4
		MCDB	37.3	39.2	44.9	56.4	67.0	69.5	72.5	69.7	61.8	52.1	44.4	40.2
	5%	WB	33.9	34.1	37.1	44.0	53.1	57.6	62.2	61.0	54.5	47.4	40.5	36.0
		MCDB	35.5	37.0	41.8	52.1	62.3	67.5	70.2	67.1	59.2	49.7	41.4	37.7
	10%	WB	31.5	32.2	35.1	41.5	50.6	55.7	60.6	59.0	52.8	45.3	38.2	33.8
		MCDB	33.0	34.4	39.1	47.9	59.2	64.9	68.1	64.2	56.5	47.4	39.0	35.4
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.4	11.7	13.9	15.6	16.2	15.4	14.0	12.5	9.5	7.2	8.2
		MCDBR	9.2	10.2	14.5	20.2	22.7	22.2	20.4	18.2	16.3	12.7	9.6	10.3
		MCWBR	8.3	7.8	9.6	11.1	11.9	10.9	9.2	9.0	9.5	8.9	8.6	9.6
	5% WB	MCDBR	8.7	10.0	13.6	18.6	20.0	19.7	17.6	15.6	15.1	11.0	8.9	10.0
		MCWBR	8.3	8.1	9.6	10.7	11.3	10.3	9.0	8.9	9.3	8.7	8.4	9.5
Clear-Sky Solar Irradiance	taub	0.222	0.273	0.299	0.333	0.338	0.333	0.349	0.341	0.310	0.296	0.240	0.341	
	taud	2.152	2.294	2.361	2.413	2.462	2.500	2.465	2.480	2.556	2.490	2.222	2.766	
	Ebn at Noon	163	226	262	273	281	284	275	268	259	220	152	113	
	Edh at Noon	11	19	25	30	31	30	30	28	21	16	10	7	
All-Sky Solar Radiation	RadAvg	82	296	798	1277	1487	1685	1540	1177	761	349	104	44	
	RadStd	12	51	75	126	158	130	187	106	102	56	26	5	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page